

Virtual reality (VR)

Introduction



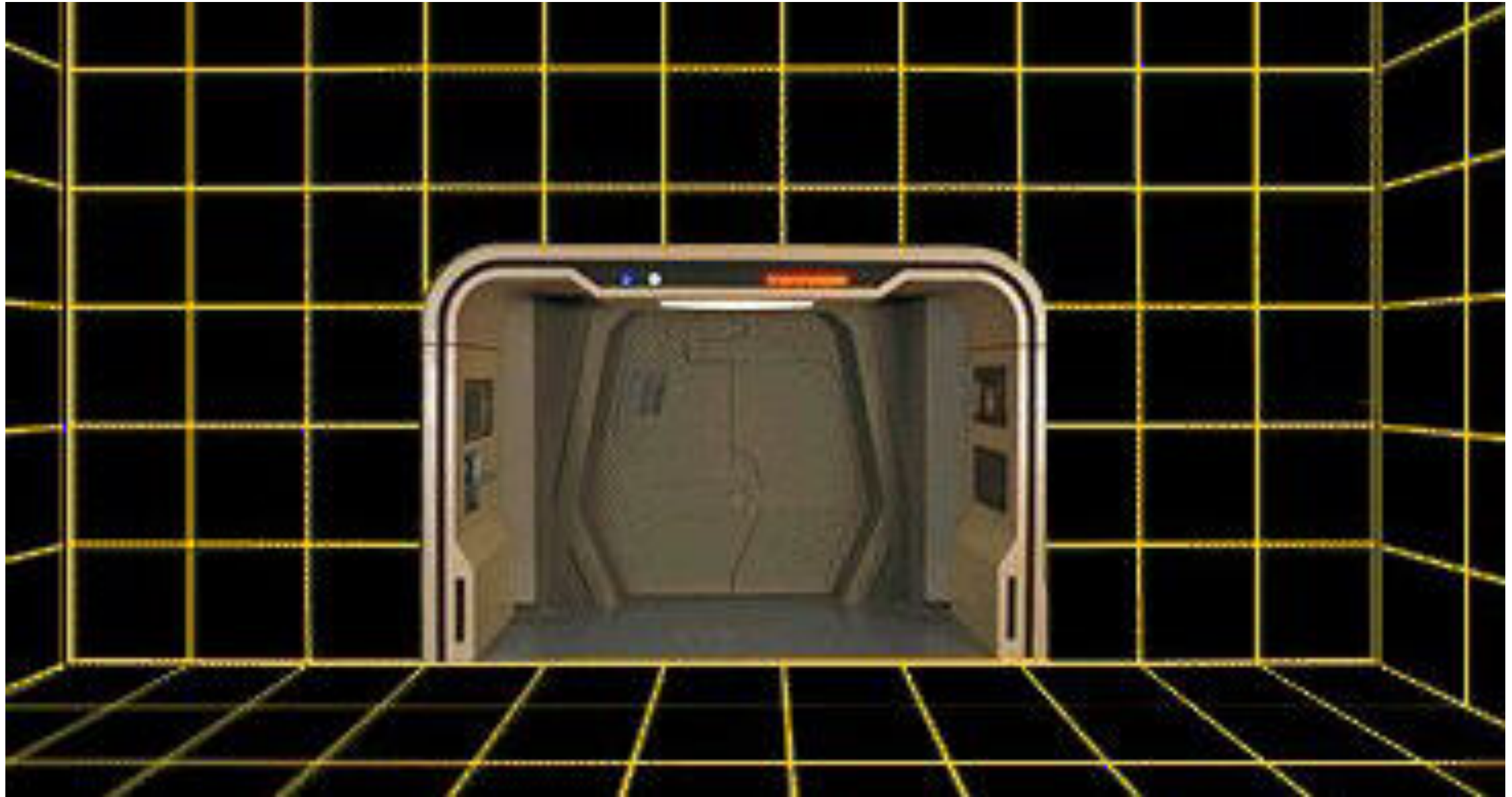
Definition - VR

- The **computer-generated simulation** of a three-dimensional image or environment that can be **interacted** with in a seemingly real or physical way by a **person** using special **electronic equipment**, such as a helmet with a screen inside or gloves fitted with sensors

Holodeck (Star Trek)



Holodeck (Star Trek)



Definition - VR

- Inducing **targeted behaviour** in an **organism** by using **artificial sensory stimulation**, while the organism has little or no **awareness** of the interference

Immersion

Low



Medium



High



Terms related to VR

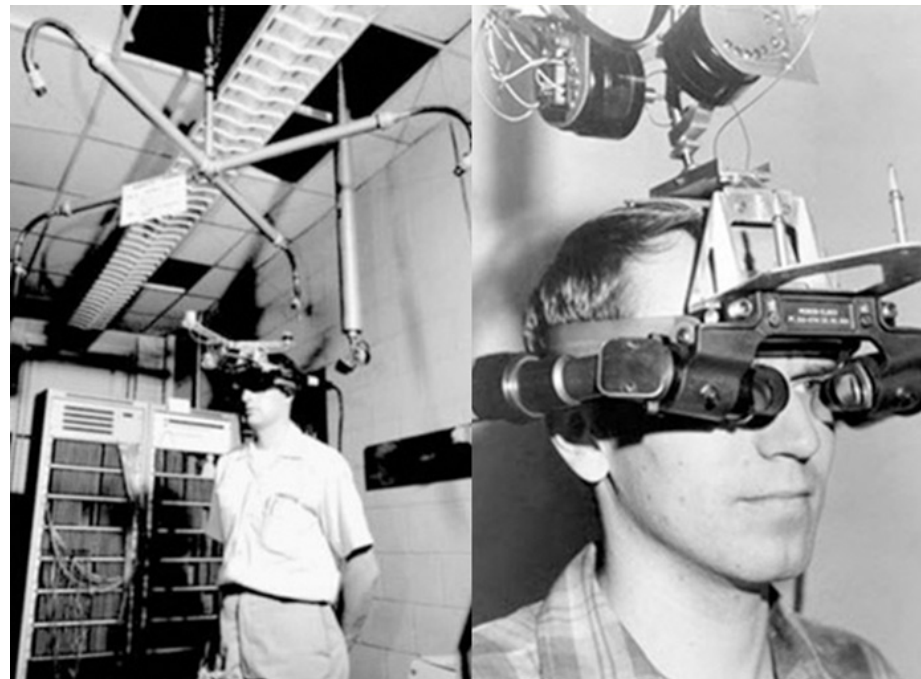
- **Virtual environments** - considered to be synonymous with VR, preferred by most university researchers
- **Telepresence** - refers to systems that enable users to feel like they are somewhere else in the real world
- **Teleoperation** - refers to systems that enable users also to control anything, such as a flying drone
- **Interactivity** - the sensory stimulation depend on actions taken by the organism
 - open-loop vs. closed-loop

History of VR

Stereoscope



VR - Ivan Sutherland



VR - Oculus Rift



1838

1968

2019

Applications

Simulation / training



Virtual surgery



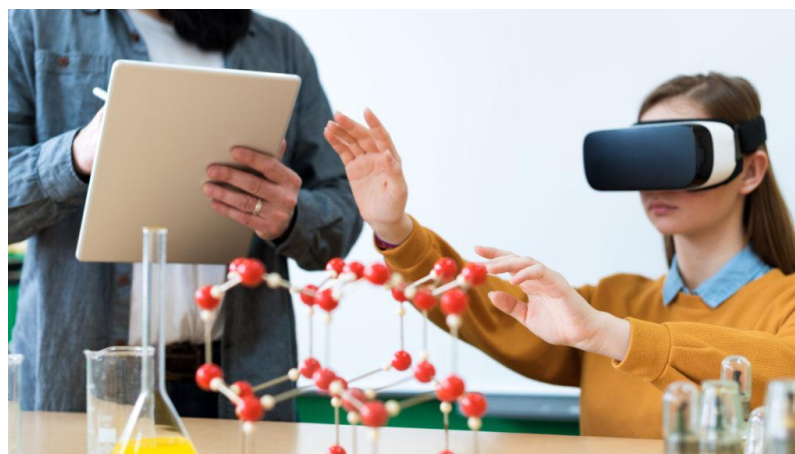
Virtual travel



Gaming



Education



Architecture



Basic headset

- Simple headset
- Uses a smartphone with no other features or functionality
- Very inexpensive
- Used for 360-degree video, also called surround video, experiences.



Mid-range headset

- Incorporates a smartphone
- Offers additional features and functionality
- Low price



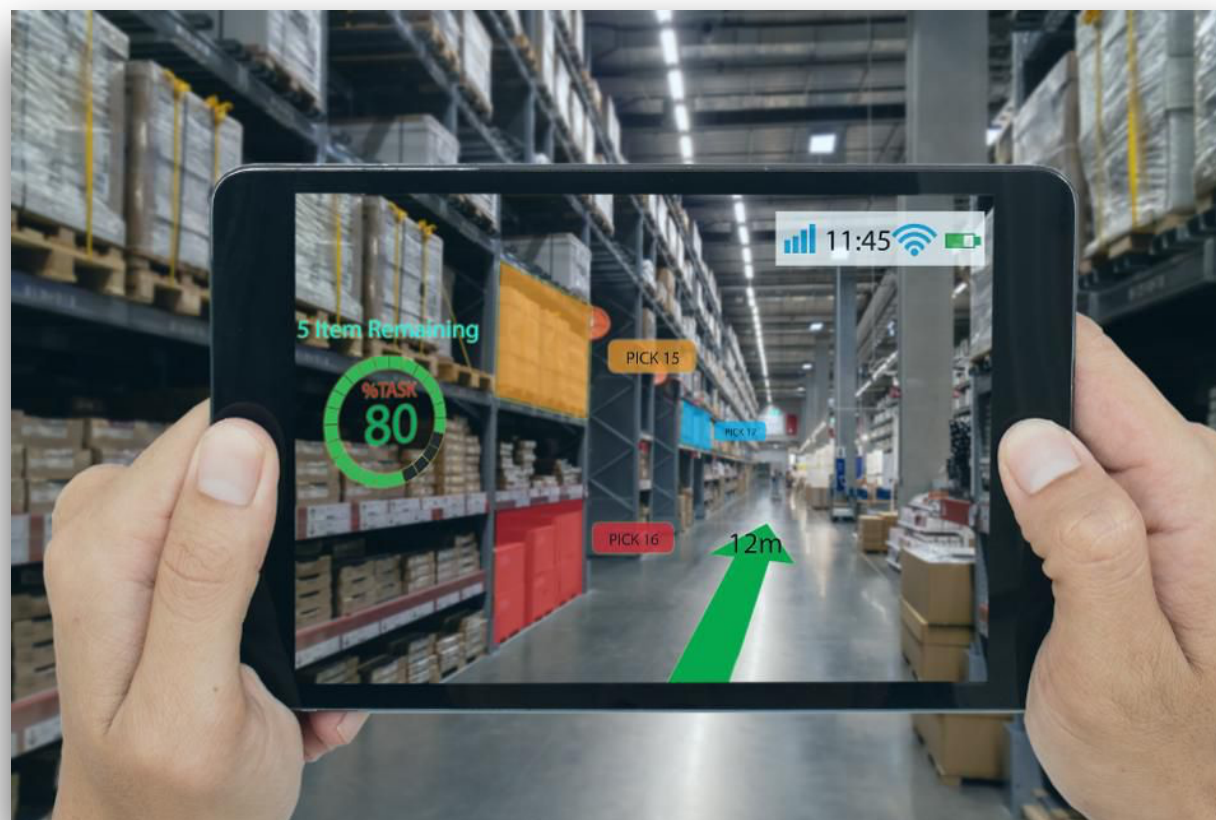
High-end headset

- Displays its own graphics
- Tracks the user's movement and behaviours
- Connected to a high-end computer capable of generating sophisticated graphics and convincing interactive experiences
- Expensive (headset, optional hand controllers, computer and high-end video card)



Augmented reality

- Superimposes a computer-generated image on a user's view of the real world
- Enhanced version of reality that uses technology to overlay digital information on an image of something being viewed through a device (such as a smartphone camera)



Mixed/merged reality

- Merging of real and virtual worlds to produce new environments and visualisations where physical and digital objects co-exist and interact in real time



Key differences

	Virtual reality (VR)	Augmented reality (AR)	Mixed/merged reality (MR)
Display device	Special headset or smart glasses	Headsets optional	Headsets optional
Image source	Computer graphics or real images produced by a computer	Combination of computer-generated images and real-life objects	Combination of computer-generated images and real-life objects
Environment	Fully digital	Both virtual and real-life objects are seamlessly blended	Both virtual and real-life objects are seamlessly blended
Perspective	Virtual objects will change their position and size according to the user's perspective in the virtual world	Virtual objects behave based on user's perspective in the real world	Virtual objects behave based on user's perspective in the real world
Presence	Feeling of being transported somewhere else with no sense of the real world	Feeling of still being in the real world, but with new elements and objects superimposed	Feeling of still being in the real world, but with new elements and objects superimposed
Awareness	Perfectly rendered virtual objects can't be distinguished from the real deal	Virtual objects can be identified based on their nature and behaviour, such as floating text that follows a user	Perfectly rendered virtual objects can't be distinguished from the real deal

Comparison of HCI styles

